Seth Balkrishan Memorial

BHILAI INSTITUTE OF TECHNOLOGY, DURG

(An Autonomous Institute affiliated to CSVTU, Bhilai)

SCHEME OF TEACHING AND EXAMINATION (Effective from 2020-2021 Batch)

B.Tech. (Information Technology) Fifth Semester

Sl. No.	Board of Studies (BOS)	Courses (Subject)	Course Code	Period per Week			Scheme of Examination			Total Marks	Credit
							Theory/Lab				
				L	T	P	ESE	CT	TA		
1.	Information Technology	Fundamental Concepts of Java Programming	106501IT	2	1	-	100	20	30	150	3
2.	Information Technology	Database Management System	106502IT	2	1	-	100	20	30	150	3
3.	Information Technology	Compiler Design	106503IT	3	-	-	100	20	30	150	3
4.	Information Technology	Software Engineering	106504IT	3	-	-	100	20	30	150	3
5.	Information Technology	Computer Networks	106505IT	3	-	-	100	20	30	150	3
6.	Information Technology	Programming in Java Lab	106591IT	-	-	2	25	-	25	50	1
7.	Information Technology	Database Management System Lab	106592IT	-	-	2	25	-	25	50	1
8.	Information Technology	Computer Networks Lab	106593IT	-	-	2	25	-	25	50	1
9.	Information Technology	Minor Project - I (JAVA)	106594IT	-	-	2	25	-	25	50	1
10.	Information Technology	Internship Assessment (Report & Seminar)	106595IT	-	-	2	-	-	25	25	1
11.	Humanities	Constitution of India	100596HM	-	-	-	-	-	25	25	-
Total				**13**	**2**	**10**	**600**	**100**	**300**	**1000**	**20**

Note:

(a) Abbreviations used : L-Lecture, T- Tutorial, P- Practical, ESE- End Semester Exam, CT- Class Test, TA- Teacher's Assessment.

(b) The duration of end semester examination of all theory papers will be of three hours.

(c) Constitution of India will be conducted by the Humanities / relevant discipline as decided by the Principal.

(d) A candidate shall have to mandatorily undertake Skill Development Courses / Internship/ Vocational Training in Industry or appropriate work place / academic and research institutions in India/Abroad, of twelve weeks duration (can be completed in parts) during the entire duration of the programme and the outcome of this should be presented in the form of a report.

SYLLABUS

B.TECH. (INFORMATION TECHNOLOGY)

FIFTH SEMESTER

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SYLLABUS

B.Tech. (Information Technology) Fifth Semester

FUNDAMENTAL CONCEPTS OF JAVA PROGRAMMING					
Course Code	**106501IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The students are made to understand the basic concepts and fundamentals of platform independent object-oriented language and use Java Standard API library functions such as util, I/O, applets, swings, GUI based controls to design and program stand-alone Java applications.	**On successful completion of the course, the student will be able to:** **CO1:** Explain the basic concepts of Java Programming. **CO2**: Apply the concepts of classes and objects in Java programs. **CO3:** Implement exception handling and String class features in Java programs. **CO4:** Create programs using Java for real-time scenarios using multithreading and File Handling. **CO5:** Design and develop application programs using UI components and Database connectivity.

UNIT I:INTRODUCTION (CO1)

History of Java, Features of Java, data types, variables, literals, expressions, operators, programming constructs, Comparison with C++, Java program Compilation and Execution, JVM, JDK, JRE, JVM as an interpreter and emulator, class File Format, Class loaders, Verification, Class Area, Java Stack, Heap, Security Promises of the JVM, Garbage Collection. **[7 Hrs]**

UNIT II: CLASSES AND OBJECTS (CO2)

Classes, Objects and References, this keyword, Methods, Accessors and Mutators, overloading method, static keyword, Access specifiers, Arrays, Enhanced for loop. Constructors and finalizers, overloading constructors, Inner classes, Wrapper Classes, Type compatibility and conversion. Inheritance; definition and advantages, overriding, Super, final and abstract classes, Interface, Package: Defining package, interfaces in package, importing packages. **[8 Hrs]**

UNIT III: EXCEPTION HANDLING, STRINGS AND COLLECTION (CO3)

Basics of exception handling, Checked and Unchecked Exceptions, default Exception handling, try and catch, Multiple catch statements, try-catch finally, uses of throw and throws, Strings: string constructor, string arithmetic, string methods, String Buffer and methods. Collections: Overview of Java collection frame work, Legacy collection classes, Commonly used collection classes and Interfaces. **[7 Hrs]**

UNIT IV: MULTI THREADING AND FILE HANDLING (CO4)

Thread Concepts, Thread life cycle, Runnable Vs Thread Class, Thread Priority, Thread Methods, Thread Synchronization: Concept of Monitor, Synchronized methods & Synchronized blocks, Deadlocks. File Handling using Java, Streams, Byte and Character Streams, Various operations with files. **[8 Hrs]**

UNIT V: GUI APPLICATION AND APPLET DEVELOPMENT (CO5)

Overview of AWT , applets and application, applet life cycle, User interfacing components, Layout Managers, Event Driven programming in java, Event delegation model, Event types and classes, Listeners, Overview of Swing Components. Introduction to JDBC, ODBC, JDBC drivers: Type I, Type II, Type III, Type IV. JDBC Architecture, executing DDL, DML,DCL commands. **[6 Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Java The complete reference	Herbert Schildt	McGraw-Hill
2.	Introduction to Java Programming	Y.Daniel Liang	Pearson Education

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Programming with JAVA	Balguruswamy	TMH
2.	Head first Java	Kathy Sierra, Bert Bates	O'Reilly Media Publication
3.	Big Java	Horstman	Willey India
4.	Effective Java	Joshua Bloch	Addison-Wesley Professional
5.	Java How to Program	H.M. Deitel & Paul J. Deitel	PHI

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SYLLABUS

B.Tech. (Information Technology) Fifth Semester

DATABASE MANAGEMENT SYSTEM					
Course Code	**106502IT**	**L = 2**	**T = 1**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The course objective is to make the students understand the basic database concepts, applications, data models, schemas and instances and entity Relationship model for a database and to Demonstrate the use of constraints and relational algebra operations and describe the basics of SQL and construct queries using SQL.	**On successful completion of the course, the student will be able to:** **CO1:** Implement the basic concepts of Database Systems in Database design and Design a Database using ER Modelling **CO2:** Construct SQL queries to interact with Database. **CO3:** Apply normalization on database design to eliminate anomalies. **CO4:** Demonstrate an understanding of database transactions and can control them by applying ACID properties. **CO5:** Apply the fundamental concepts of transaction processing- concurrency control techniques and recovery procedure.

UNIT I: **[CO1]**

Introduction: Introduction and applications of DBMS, Purpose of data base, Data, Independence, Database System architecture- Levels, Mappings, Database, users and DBA.

Database design: Database Design Process, ER Diagrams - Entities, Attributes, Relationships, Constraints, keys, extended ER features, Generalization, Specialization, Aggregation, Conceptual design with the E-R model.

Structure of Files: Types of Single Level ordered indexes, Multilevel indexes, Dynamics Multilevel indexes using B and B^+ Tree. **[8 Hrs]**

UNIT II: **[CO2]**

Relational Algebra & Calculus: Preliminaries, relational algebra operators, relational calculus - Tuple and domain relational calculus, expressive power of algebra and calculus.

The Relational Data Model: Relational data model concepts, constraints, relational algebra, relational calculus, SQL: DDL, DML, DCL, View, Index, Cursors and Triggers. **[7 Hrs]**

UNIT III: **[CO4]**

Schema Refinement & Normal Forms: Functional Dependency – definition, trivial and non-trivial FD, closure of FD set, closure of attributes, irreducible set of FD, Normalization – 1Nf, 2NF, 3NF, Decomposition using FD- dependency preservation, BCNF, Multivalued dependency, 4NF, Join dependency and 5NF. **[7 Hrs]**

UNIT IV: **[CO4]**

Query & Transaction Processing: Query Processing: Query processing stages.

Transaction: Types of failures, ACID property, schedules and recoverability, serialisability of schedules, Levels of transaction consistency, Deadlocks, Nested transaction. **[7 Hrs]**

		August 2022	1.00	Applicable for
Chairman (AC)	Chairman (BoS)	Date of Release	Version	AY 2022-23 Onwards

UNIT V: **[CO5]**

Transaction Management & Crash Recovery: Failure classification, Different type of Recovery techniques & their comparative analysis, deferred update, immediate update, Shadow paging, Check points, On-line backup during database updates, Concurrency Control: Different type of concurrency control techniques & their comparative analysis, Locking techniques, Time-stamp ordering, Multi-version techniques, Optimistic techniques, Multiple granularity, Integrity, Security, Non-procedural procedural integrity constraints and Integrity constraints specifications in SQL. **[7 Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Database System Concepts	Korth & Sudarshan	Tata McGraw Hill
2.	Database Management Systems	Raghurama Krishnan, Johannes Gehrke	Tata McGraw Hill
3.	Introduction to Database Systems	C. J. Date	Pearson Education

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Principles of Database Systems	Ullman, J.O.	Galgotia Publications
2.	Fundamentals of Database Systems	Elmasri R. & Navathe S.B	Pearson Education

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SYLLABUS

B.Tech. (Information Technology) Fifth Semester

COMPILER DESIGN					
Course Code	**106503IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The goal of the course is to provide an introduction to the system software like assemblers, compilers, and macros. It provides the complete description about inner working of a compiler. This course focuses mainly on the design of compilers and optimization techniques. It also includes the design of Compiler writing tools. This course also aims to convey the language specifications, Code generation, machine dependent code optimization, use of regular expressions and context free grammars behind the design of compiler.	**On successful completion of the course, the students will be able to:** **CO1:** Identify the relationships among different phases of the compiler. **CO2:** Determine the Types and roles of the lexical analyzers. **CO3:** Apply the concepts of different Parsing techniques and implement the knowledge to YACC tool. **CO4:** Use the new code optimization techniques to improve the performance of a program in terms of speed & space and develop algorithms to generate code for a target machine. **CO5:** Illustrate the various optimization techniques for designing various optimizing compilers

UNIT I: Introduction **(CO1)**

Introduction to Compiler, single and multi-pass compilers, Translators, Phases of Compilers, Compiler writing tools, Finite Automata and Lexical Analyzer: Role of Lexical Analyzer, Specification of tokens, Recognition of tokens, Regular expression, Finite automata, from regular expression to finite automata, transition diagrams, Implementation of lexical analyzer with LEX. **[7 Hrs]**

UNIT II: Syntax Analysis and Parsing Techniques **(CO2)**

Context free grammars, Bottom-up parsing and top down parsing, Top down Parsing: elimination of left recursion, recursive descent parsing, Predicative Parsing, Bottom Up Parsing: Operator precedence parsing, LR parsers, Construction of SLR, Canonical LR and LALR parsing tables, Construction of SLR parse tables for Ambiguous grammar, parser generator- YACC, error recovery in top down and bottom up parsing. **[8 Hrs]**

UNIT III: Syntax Directed Translation & Intermediate Code **(CO3)**

Synthesized and inherited attributes, Construction of syntax trees, bottom up and top down evaluation of attributes, S- attributed and L-attributed definitions, Postfix notation; Three address codes, quadruples, triples and indirect triples, Translation of assignment statements, control flow, Boolean expression and Procedure Calls. **[8 Hrs]**

UNIT IV: Run-time Environment **(CO4)**

Storage organization, activation trees, activation records, allocation strategies, Parameter passing symbol table, dynamic storage allocation. **[7 Hrs]**

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UNIT V: Code Optimization and Code Generation **(CO5)**

Basic blocks and flow graphs, Optimization of basic blocks, Loop optimization, and global data flow analysis, Loop invariant computations. Issue in the design of Code generator, register allocation, the target machine, and simple Code generator. **[6 Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Compilers Principles, Techniques and Tools	Alfred V. Aho, Ravi Sethi & Ullman J.D	Addison Wesley.
2.	Principles of compiler design	Alfred V. Aho, Jeffrey D. Ullman	Pearson Education
3.	Introduction to Compiler Techniques	J.P. Bennet	Tata McGraw-Hill

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Compiler Design in C	A.C. Holub	PHI
2.	Modern Compiler Implementation C	Andrew W. Appel	Cambridge University Press
3	Compiler Design	O.G. Kakde	Laxmi Publication
4	Crafting a Compiler with C	Charles Fischer and Ricard LeBlanc	Pearson Education

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B.Tech. (Information Technology) Fifth Semester

SOFTWARE ENGINEERING					
Course Code	**106504 IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The course objective is to make the students to learn and understand the principles of Software Engineering. To be acquainted with methods of capturing, specifying, visualizing and analyzing software requirements. To apply Design and Testing principles to Software project development. To understand software quality attributes.	**On successful completion of the course, the student will be able to:** **CO1:** Select a process model for developing a software project **CO2:** Classify software applications and Identify unique features of various domains **CO3:** Design test cases for different types of software system testing. **CO4:** Apply different phases of software Project management **CO5:** Analyze risks and estimate the cost factors for implementing the projects.

UNIT I: Introduction to Software Engineering, Software Process Models **(CO1)**

Introduction to software engineering- scope of software engineering – historical aspects, economic aspects, maintenance aspects, specification and design aspects, team programming aspects. The Software Process, Software Myths. Process Models: A Generic Process Model, The Waterfall, Incremental Process. RAD, Prototyping, Evolutionary Process, Object oriented model. **[7 Hrs]**

UNIT II: Software Requirements Engineering & Analysis **(CO2)**

Requirements Engineering: User and system requirements, Functional and non-functional requirements, Types & Metrics, A spiral view of the requirements engineering process. Software Requirements Specification (SRS): The software requirements Specification document, requirements elicitation for software, analysis principles, software prototyping, specification. **[9 Hrs]**

UNIT III: Design Engineering **(CO3)**

Design Process & quality, Design Concepts, the design Model, Architectural Design, Modeling Component Level Design, User Interface Design, effective modular design, top down, bottom up strategies, DFD, Data Dictionary, HIPO Chart, Use case diagram. **[8 Hrs]**

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UNIT IV: Software Quality Assurance and Testing **(CO4)**

Testing Objectives, Testing Principles, Test Plans, Test Cases, Types of Testing, Levels of Testing, Test Strategies, Program Correctness, Program Verification & validation, Testing Automation & Testing Tools, Concept of Software Quality, Software Quality Attributes, Software Quality Metrics and Indicators, The SEI Capability Maturity Model CMM), SQA Activities, Formal SQA Approaches: Proof of correctness, Statistical quality assurance, Clean room process. **[6 Hrs]**

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UNIT V: Software Project Management **(CO5)**

Project Management Concepts**:** The Management Spectrum, People, Product, Process, Project, Metrics and Measurement: size & function oriented metrics (FP & LOC), Metrics for Project and Software Quality, Project Estimation Software Project Estimation, Decomposition Techniques, Empirical Estimation Models: Structure, COCOMO, Estimation of Object-oriented Projects, Specialized Estimation. Risk Management Process: Risk identification, Risk analysis, Risk planning, Risk monitoring. Maintenance & Reengineering: Software Maintenance, Software Supportability, Reengineering, Business Process Reengineering, Software Reengineering. **[6 Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Software Engineering: A Practitioner ‘s approach	Roger Pressman	McGraw-Hill
2.	Project Management: The Managerial Process with MS	Clifford F. Gray and Erik W. Larson	McGraw-Hill

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Fundamentals of Software Engineering	Carlo Ghezzi	Prentice Hall India
2.	Fundamentals of Software Engineering	Rajib Mall	Prentice Hall India
3.	An Integrated Approach to Software Engineering	Pankaj Jalote	Springer

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B.Tech. (Information Technology) Fifth Semester

COMPUTER NETWORKS					
Course Code	**106505IT**	**L = 3**	**T = 0**	**P = 0**	**Credits = 3**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	100	**20**	**30**	**150**	**3 Hours**
	Minimum number of class tests to be conducted = 02			**Minimum assignments = 02**	

Course Objectives	Course Outcomes
The course introduces main concepts of networking; application areas; classification; reference models; transmission environment; technologies; routing algorithms; IP, UDP and TCP protocols; reliable data transferring methods; application protocols; network security; management systems; perspectives of communication networks.	**On successful completion of the course, the student will be able to:** **CO1:** Demonstrate an understanding of the fundamental concepts of computer networking. **CO2**: Define the basic taxonomy and terminology of the computer networking area. **CO3:** Identify the different types of network topologies and protocols. **CO4:** Demonstrate an understanding of the client/server model and key application layer protocols. **CO5:** Enumerate the layers of the OSI model and TCP/IP and understand the basics of error detection including parity, checksums, and CRC.

UNIT – I INTRODUCTION **(CO1)**

Introduction: Data communications, networks, network types. **Network Models:** Protocol layering, TCP/IP protocol suite, The OSI model. **Transmission media:** Guided Media, Unguided Media. **Switching:** Introduction, circuit switched networks, packet switching, structure of a switch, Telephone network. **[7 Hrs]**

UNIT - II DATA LINK LAYER: **(CO2)**

DATA LINK LAYER : Data link Layer design issues, Elementary Datalink Protocols, Error Detection and Correction- Hamming code, CRC, Parity, Checksum, Switch working, HDLC.

MAC SUBLAYER: Multiple Access Protocols- ALOHA, CSMA and CSMA/CD, Random Access, Controlled access, Ethernet protocol. **[8 Hrs]**

UNIT - III NETWORK LAYER: **(CO3)**

NETWORK LAYER: IP Addressing : Network layer design issue, IP Addressing Both Classful and Classless, Subnetting and Supernetting, Subnetting examples, Network layer services, Network layer performance, Forwarding of IP packets, IP Header, IPv6 addressing

Routing : Routing Algorithm-Shortest path algorithm, Distance vector Routing, Link State routing, Routing algorithms, Unicast routing protocols **[9 Hrs]**

UNIT – IV TRANSPORT LAYER: **(CO4)**

Transport service, elements of transport protocol, Simple Transport Protocol, Internet transport layer protocols: UDP and TCP, data traffic, congestion, congestion control, QoS, integrated services, differentiated services, QoS in switched networks. **[6 Hrs]**

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UNIT - V APPLICATION LAYER: **(CO5)**

Domain name system, electronic mail, World Wide Web: architectural overview, dynamic web document and http. APPLICATION LAYER PROTOCOLS: Simple Network Management Protocol, File Transfer Protocol, Simple Mail Transfer Protocol, Telnet. **[6 Hrs]**

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Computer Networks	S. Tanenbaum	Pearson Education
2.	Data communication and Networking	Behrouz A. Forouzan	McGraw Hill Education
3.	Data and Computer Communications	Walliam Stallings	PHI

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Computer Networking: A top down approach,	Kurose, Ross	Pearson Education
2.	Computer Networks: Protocols standards and interfaces	Uyless Black,	Prentice Hall
3	Computer Network	V. S. Bagad, I. A. Dhotre	Technical Publication

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B.Tech. (Information Technology) Fifth Semester

PROGRAMMING IN JAVA LAB					
Course Code	**106591IT**	**L = 0**	**T = 0**	**P = 2**	**Credits = 1**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	25	**-**	**25**	**50**	**-**

Course Objectives	Course Outcomes
The course objective is to make the students understand the basic concepts and fundamentals of platform independent object oriented language. To use Java Standard API library such as util, io, applets, swings, GUI based controls. To learn java's exception handling mechanism, multithreading, packages and interfaces. To design and program stand-alone Java applications.	**On successful completion of the course, the student will be able to:** **CO1:** Develop Java programs using class, objects and constructors. **CO2**: Utilize the concept of inheritance to develop Java programs. **CO3:** Demonstrate the use of exception handling and Strings in java programs. **CO4:** Create multithreaded applications using Java programming. **CO5:** Design and develop interactive application programs using user Interfacing components, file handling, and JDBC.

1. Write a Java Program to define a class, define instance methods for setting and Retrieving values of instance variables and instantiate its object.

2. Write a program to find the volume of a box having its side w,h,d means width ,height and depth.

Its volume is v=w*h*d and also find the surface area given by the formula s=2(wh+hd+dw). use

appropriate constructors for the above.

3. Create a base class called shape. Apart from Constructors, It contains two methods get xyvalue()

and show xyvalue() for accepting co-ordinates and to display the same. Create the sub class Called

Rectangle which contains a method to display the length and breadth of the rectangle called

showxyvalue().Illustrate the concepts of Overriding and Constructor call sequence.

4. Write a program in JAVA to create a class Bird also declares the different parameterized constructor to display the name of Birds.

5. Write a program that creates an abstract class called dimension, create two subclasses, rectangle

and triangle. Include appropriate methods for both the subclass that calculate and display the area

of the rectangle and triangle.

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6. Write a program, which throws Arithmetic Exception. Write another class (in a different file) that handles the Exception.

7. Create a user defined Exception class which throws Exception when the user inputs the marks greater than 100 Catch it and again rethrow it.

8. Write a program to illustrate various String class methods.

9. Write a program to illustrate various String Buffer methods.

10. Write a Java program to illustrate collection classes.

11. Write a program in which a Mythread class is created by extending the Thread class. In another class, create objects of the Mythread class and run them. In the run method print "Java" 10 times. Identify each thread by setting the name.

12. Write a program to illustrate various Thread methods.

13. Write a Program to implement Bank Account Class which illustrates the concept of Thread Synchronization.

14. To write a program to create a text file using Byte Stream class.

15. To write a program to copy contents of one file to another.

16. Write a java program to read a file and display number of vowels, consonants, digits and special characters present in file.

17. Write a program, which illustrates capturing of Mouse Events. Use Applet for this.

18. Write a program using swing components which simulates simple calculator.

19. Write a JDBC program for Student Mark List Processing.

20. Design a text editor, which is having some of the features of notepad.

Laboratory Equipments / Machine Requirements: Windows / Linux operating system, 4GB RAM, 5GB free disk space and Jdk 1.5 or higher version

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Java The complete reference	Herbert Schildt	McGraw-Hill
2.	Introduction to Java Programming	Y.Daniel Liang	Pearson Education

Reference Books:

S.No.	Title	Author(s)	Publisher
1.	Programming with JAVA	Balguruswamy	TMH
2.	Head first Java	Kathy Sierra, Bert Bates	O'Reilly Media Publication
3.	Big Java	Horstman	Willey India
4.	Effective Java	Joshua Bloch	Addison-Wesley Professional
5.	Java How to Program	H.M. Deitel & Paul J. Deitel	PHI

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B.Tech. (Information Technology) Fifth Semester

DATABASE MANAGEMENT SYSTEM LAB					
Course Code	**106592IT**	**L = 0**	**T = 0**	**P = 2**	**Credits = 1**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	25	**-**	**25**	**50**	**-**

Course Objectives	Course Outcomes
The course objective is to make the students understand the features of a Database Management Systems and to interface databases with front end tools.	**On successful completion of the course, the students will be able to:** **CO1:** Create and maintain databases and tables. **CO2:** Design and implement a database schema for a given problem-domain. **CO3:** Populate and query a database using SQL DML/DDL commands. **CO4:** Create PL/SQL programs using stored procedures, stored functions, cursors, packages. **CO5:** Manage transactions and locks to ensure data concurrency and recoverability.

List of Experiments (but should not be limited to):

1. Creating and Manipulating Database objects and Applying Constraints (DDL)
2. Manipulating Data with Database Objects (DML).
3. Retrieving, Restricting and Sorting Data (DRL)
4. Apply the concept of Aggregating Data using Group functions.
5. Displaying data from Multiple Tables (join)
6. Solve queries using the concept of sub query.
7. Set operators & Views in SQL
8. Control structures.
9. Working with Procedures and Functions.
10. Triggers
11. Study Transaction control commands.

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Database Management Systems (DBMS)	Rajiv Chopra	S. Chand
2.	The Complete reference (SQL)	James R. Groff	Mc Graw Hill
3.	Sams Teach Yourself SQL	Ben Forta	Sams Publishing

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B.Tech. (Information Technology) Fifth Semester

COMPUTER NETWORKS LAB					
Course Code	**106593IT**	**L = 0**	**T = 0**	**P = 2**	**Credits = 1**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	25	**-**	**25**	**50**	**-**

Course Objectives	Course Outcomes
The course objective is to make students familiar with simulation tools as well as various routing algorithms and to provide hands on experience on various networking protocols for communication between two desktop computers.	**On successful completion of the course, the student will be able to:** **CO1:** Describe the general principles of data communication. **CO2**: Use simulation tools **CO3**: Implement the various protocols. **CO4:** Analyze the performance of the protocols in different layers. **CO5:** Analyze various routing algorithms

List of Experiments

1. To learn handling configuration and testing of networking hardware like RJ-45 connector, CAT-6 cable, crimping tool and LAN tester.

2. Study of complete network architecture of your institution (including topology, network devices cabling standards, protocol and security features).

3. Test the connectivity between two hosts.

4. Write a Program to find the IP address and domain name of your system.

5. Make a report on LAN establishment in any of organization including hardware & networking aspects.

6. To learn configuration of router, hub, switch in a network. (Using real devices or simulators).

7. Running and using services/commands like ping, trace route, ns lookup, arp, telnet, ftp, etc.

8. To get the MAC or Physical address of the system using Address Resolution Protocol.

9. Simulation of Star, Bus, Ring Topology Using LAN Trainer Kit. 8. Study the different Network Standard Topologies.

10. Installation and working of Net meeting and Remote Desktop.

11. To study Client-Server Communication Using Switch.

12. Write a program for a HLDC frame to perform the following.

a. Bit stuffing.

b. Character stuffing.

13. Implementation of Error Detection / Error Correction Techniques. Verify the program for the cases i. Without error ii. With error

14. Implementation of Stop and Wait Protocol and Sliding Window Protocol.

15. Implementation and study of Goback-N and selective repeat protocols.

16. Perform a case study about the different routing algorithms to select the network path with its optimum and economical during data transfer. i. Shortest path routing ii. Flooding iii. Link State

17. Create a socket (TCP) between two computers and enable file transfer between them.

18. Write a program to encrypt and decrypt a file using a key.

Text Books:

S.No.	Title	Author(s)	Publisher
1.	Computer Networks	S. Tanenbaum	Pearson Education
2.	Data communication and Networking	Behrouz A. Forouzan	McGraw Hill Education
3.	Data and Computer Communications	Walliam Stallings	PHI

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Bhilai Institute of Technology, Durg

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SYLLABUS

B.Tech. (Information Technology) Fifth Semester

MINOR PROJECT – I (JAVA)					
Course Code	**106594IT**	**L = 0**	**T = 0**	**P = 2**	**Credits = 1**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	25	**-**	**25**	**50**	**-**

Course Objectives	Course Outcomes
The Course objective is to provide an opportunity for students to understand real-life practices and implementation of theoretical Engineering knowledge in practical environment. To provide an opportunity for students to acquire interpersonal skills and ability for team work. To obtain knowledge of how to make optimal decisions to resolve technical challenges.	**On successful completion of the course, the student will be able to:** **CO1:** Apply engineering knowledge in real life projects by integrating theoretical concepts with real-time situations. **CO2:** Acquire practical skills and Communication skills while working on projects. **CO3:** Obtain knowledge of how to make optimal decisions to resolve technical challenges. **CO4:** Write technical documents and give oral presentations related to the project work completed.

- Minor project is introduced in the fifth semester with a specific objective to strengthen the understanding of student's fundamentals through effective application of theoretical concepts of JAVA Programming Language.
- Student Groups with 3 or 4 members should identify a topic of interest in consultation with Faculty Guide.
- Review the literature and gather information pertaining to the chosen topic.
- State the objectives and develop a methodology to achieve the objectives.
- Carryout the design/ develop codes/programs to achieve the objectives.
- Demonstrate the novelty of the project through the results and outputs.
- The progress of the project is evaluated based on a minimum of two reviews.
- The review committee may be constituted by the Head of the Department.
- A project report is required at the end of the semester.
- The product has to be demonstrated for its full design specifications.
- The internal evaluation will be made based on the product, the report and a viva-voce examination, conducted internally by a committee appointed by Head of the Department comprising HoD or a Project coordinator for that program.

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SYLLABUS

B.Tech. (Information Technology) Fifth Semester

INTERNSHIP ASSESSMENT (REPORT & SEMINAR)					
Course Code	**106595IT**	**L = 0**	**T = 0**	**P = 2**	**Credits = 1**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	-	**-**	**25**	**25**	**-**

Course Objectives	Course Outcomes
The course objective is to promote cooperation and to develop synergetic collaboration between industry and the institute in promoting a knowledgeable society. To expose students to the 'real' working environment and get acquainted with the organization structure, business operations and administrative functions.	**On successful completion of the course, the student will be able to:** **CO1**: Apply engineering knowledge in solving real-life problems. **CO2**: Attain new skills and be aware of the state-of-art in engineering disciplines of their own interest. **CO3**: Describe use of advanced tools and techniques encountered during training and visit. **CO4**: Acquire practical skills, organizational skills, Communication skills, lifelong learning skills, professional awareness and experience working on projects and alongside industry experts. **CO5:** Write a technical project report that follows an established structure and give oral presentations with focus on the project results and a credible work procedure.

1. As per institution's guidelines, Industrial Training have to be done at the end of 4th Semester.

2. The student must follow the instructions given by the Teacher In-charge of Industrial Training.

3. The purpose of the Industrial Training is to develop the work process being performed and apprise them of the industry problems.

4. During the training, students will be given practical problems by the industry in which they are undergoing training. In case the industry does not give them the problems, the students will themselves formulate problems and carry out detailed study on them and recommend the optimum solution based on their theory knowledge.

5. Submission of Final Report - Each student must submit a Final Report and deliver a presentation on industrial training at the end of the industrial training period on the date and time announced by the Teacher In-charge.

		August 2022	1.00	Applicable for AY 2022-23 Onwards
Chairman (AC)	Chairman (BoS)	Date of Release	Version	

6. The Internship report must include the following:

(a) The basic history/introduction of the industry.

(b) The software and hardware used.

(c) The sequence of operations followed/ systems introduced for the project development.

(d) The formulation of practical problems.

(e) Data required formulating the problems.

(f) Analysis of the data, steps required and commands used in industry.

(g) Certificate from the industry for the period of training undergone.

7. The student would be evaluated through Report and Viva-voce.

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SYLLABUS

B.Tech. (Common to All Branches) Fifth Semester

CONSTITUTION OF INDIA					
Course Code	**100596HM**	**L = 0**	**T = 0**	**P = 0**	**Credits = -**
Examination Scheme	**ESE**	**CT**	**TA**	**Total**	**ESE Duration**
	-	**-**	**25**	**25**	**-**

Course Objectives	Course Outcomes
The objective of this course is to introduce students to the Constitution of India.	**On successful completion of the course, the student will be able to:** **CO1:** Display understanding about the history and philosophy of Indian Constitution. **CO2:** Demonstrate clarity about the premises informing the twin themes of liberty and freedom from civil rights perspective. **CO3:** Display understanding about powers and functions of Indian government. **CO4:** Exhibit understanding about emergency rule. **CO5:** Demonstrate understanding about structure and functions of local administration.

UNIT – I **CO1)**

Introduction: Historical Perspective of Constitution of India; Philosophy of Indian Constitution; Meaning of the constitution law and constitutionalism; Salient features and Preamble.

UNIT – II **(CO2)**

Contours of Constitutional Rights and Duties: Fundamental rights; Scheme of the Fundamental Duties and its legal status.

UNIT – III **(CO3)**

Organs of Governance: Parliamentary Form of Government in India; The constitutional powers and status of the President of India; Judiciary- Powers and Functions; Local Self Government – Constitutional Scheme in India.

UNIT – IV **(CO4)**

Emergency Provisions: National Emergency; President Rule; Financial Emergency.

UNIT – V **(CO5)**

Local Administration: Federal structure and distribution of legislative and financial powers between the Union and the States; The Directive Principles of State Policy – Its importance and implementation.

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Text Books:

S. No.	Title	Author(s)	Publisher
1.	Introduction to the Constitution of India	Basu D D	Lexis Nexis
2.	Principles of Public Administration	Dr S N Myneni	Allahabad Law Agency

Reference Books:

S. No.	Title	Author(s)	Publisher
1.	Dr. B. R. Ambedkar Framing of Indian Constitution	Busi S N	Ava Publishers
2.	Theory and Practices of Modern Government	M G Gupta	Central Book Depot